

MECHANISM OF SECRETION OF HCl

- **Davenport Theory** :—

- HCl secretion is an **active process**.
- Takes place in the **canaliculi of parietal (oxyntic) cells** in gastric glands.
- Energy is derived from the **oxidation of glucose**.
- CO_2 is derived from the metabolic activities of the parietal cell.
- Enzyme **Carbonic Anhydrase** is present in high concentration inside parietal cells.

CELLULAR MECHANISM & FLOWCHART

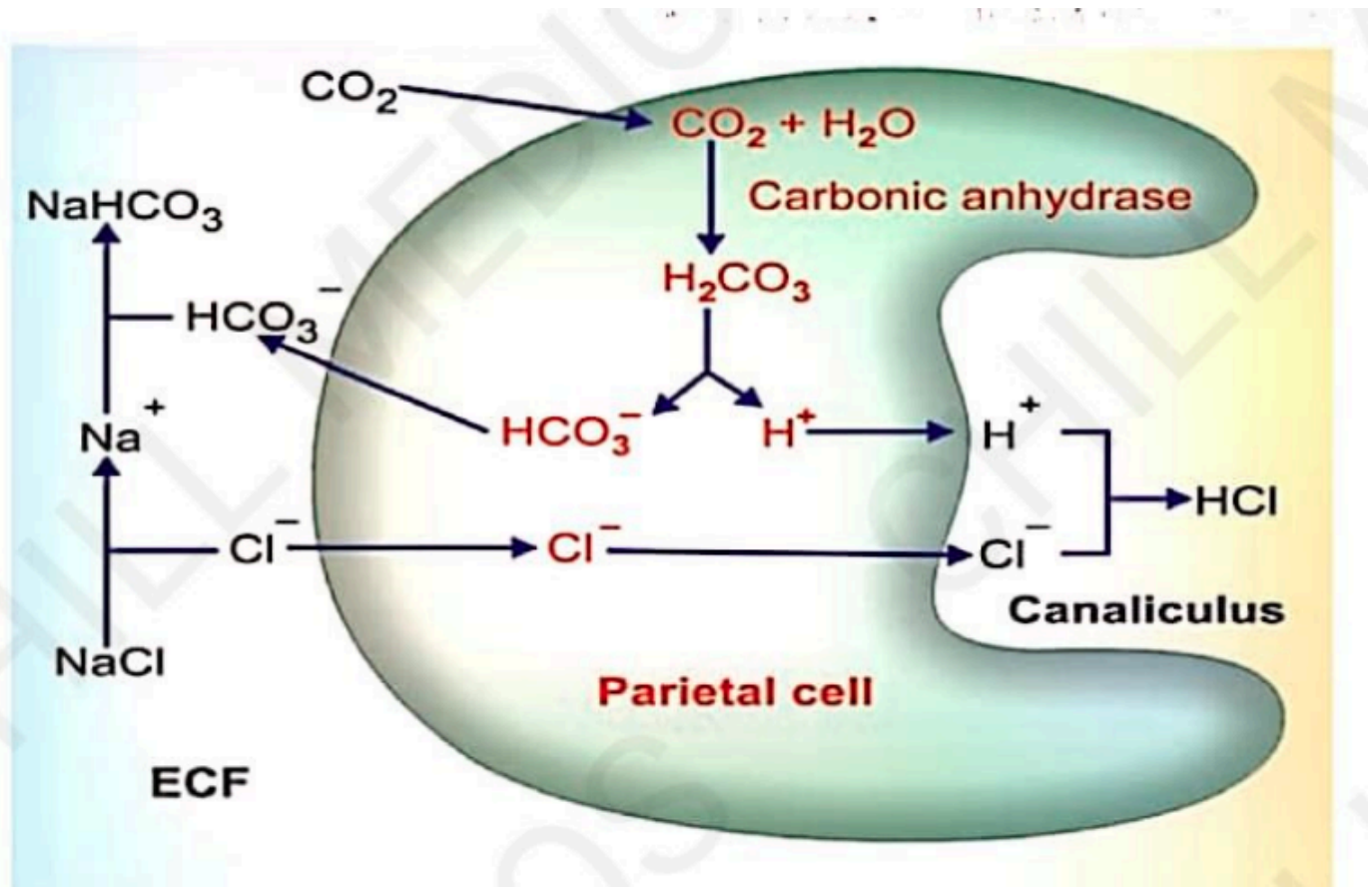
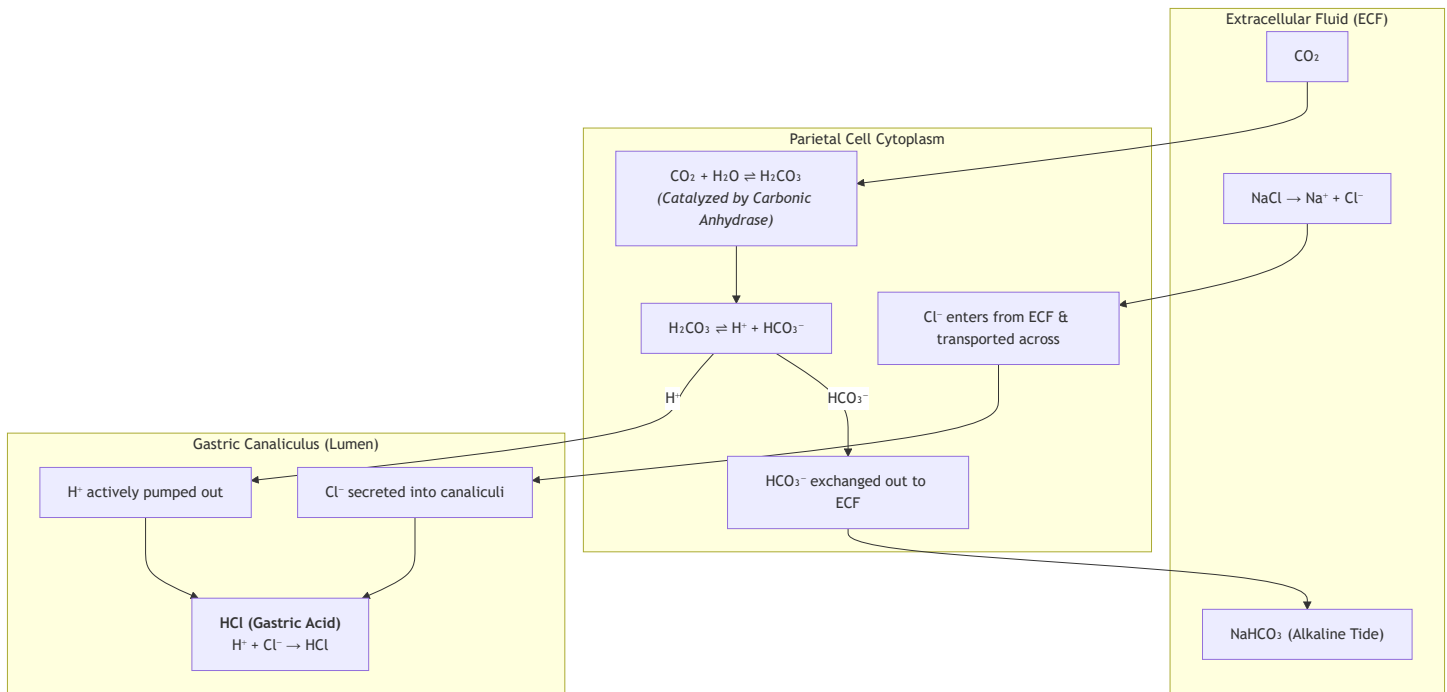
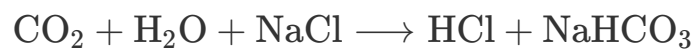


FIGURE 38.4: Secretion of hydrochloric acid in the parietal cell of gastric gland



OVERALL CHEMICAL REACTION



REGULATION OF HCl SECRETION

Stimulants of Secretion of HCl	Inhibitors of Secretion of HCl
• Gastrin • Histamine • Vagal stimulation	• Secretin • Peptide YY • Gastric Inhibitory Polypeptide (GIP)